Lab Exercise 4
EEE301: Signals and Systems

<u>Section A</u>: Using MATLAB/OCTAVE to plot the Bode plot of a transfer function.
Consider the transfer function $H(s)$ of an LTI system

$$H(s) = \frac{P(s)}{Q(s)}$$

MATLAB function `tf(num, den)` can be used to define a transfer function $H(s)$, where `num` is the list of the coefficients of numerator polynomial $P(s)$ and `den` is the list of the coefficients of the denominator $Q(s)$. For example,

$$H(s) = \frac{P(s)}{Q(s)} = \frac{s+3}{s(s+1)(s+2)} = \frac{s+3}{s^3+3s^2+2s}$$

If we want to define the transfer function $H(s)$ above in MATLAB, we write:

```
num = [1 3];
den = [1 3 2 0];
H = tf(num, den);
```

To plot the Bode plot of the transfer function $H(s)$, use the MATLAB function `bode(H, w)`, where the `H` is the transfer function returned by MATLAB function `tf(num, den)` and `w` is the list of frequencies ω to plot. For example, let us plot the transfer function $H(s)$ above for frequency defined in the range
$0.1 \leq \omega \leq 100$.

```
w = [0.1:0.5:100];
bode(H, w);
```

Run the MATLAB script above and observe the Bode plot, as shown in Figure 1 below.

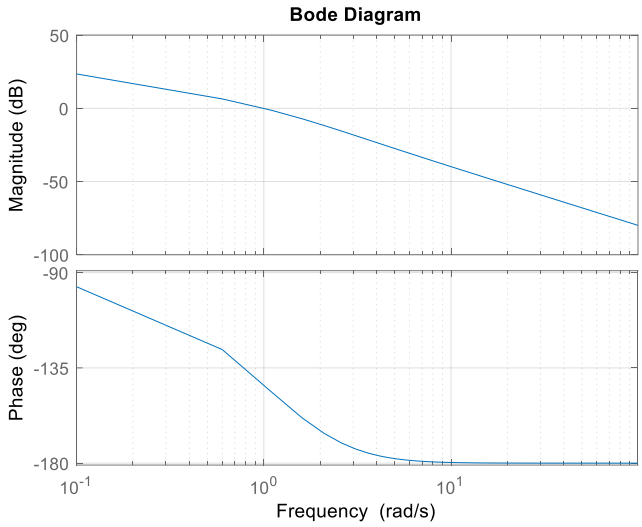


Figure 1

Exercise A

Use MATLAB to obtain the Bode plot for the following transfer functions:

(a) $H(s) = \frac{s+10}{10}$ for $0.1 \leq \omega \leq 1000$.

(b) $H(s) = \frac{100}{s+100}$ for $1 \leq \omega \leq 1000$.

(c) $H(s) = \frac{10^4(s+1)}{(s+10)(s+100)}$ for $0.1 \leq \omega \leq 1000$.

For each of the transfer function above, determine from the Bode plot, the value of the magnitude $|H(j\omega)|$ and the phase angle $\angle H(j\omega)$ at selected frequency points shown in Table 1 below.

ω	$\|H(j\omega)\|$ (dB)	$\angle H(j\omega)$ (deg)
1		
10		
20		
50		
100		
500		
1000		

Table 1

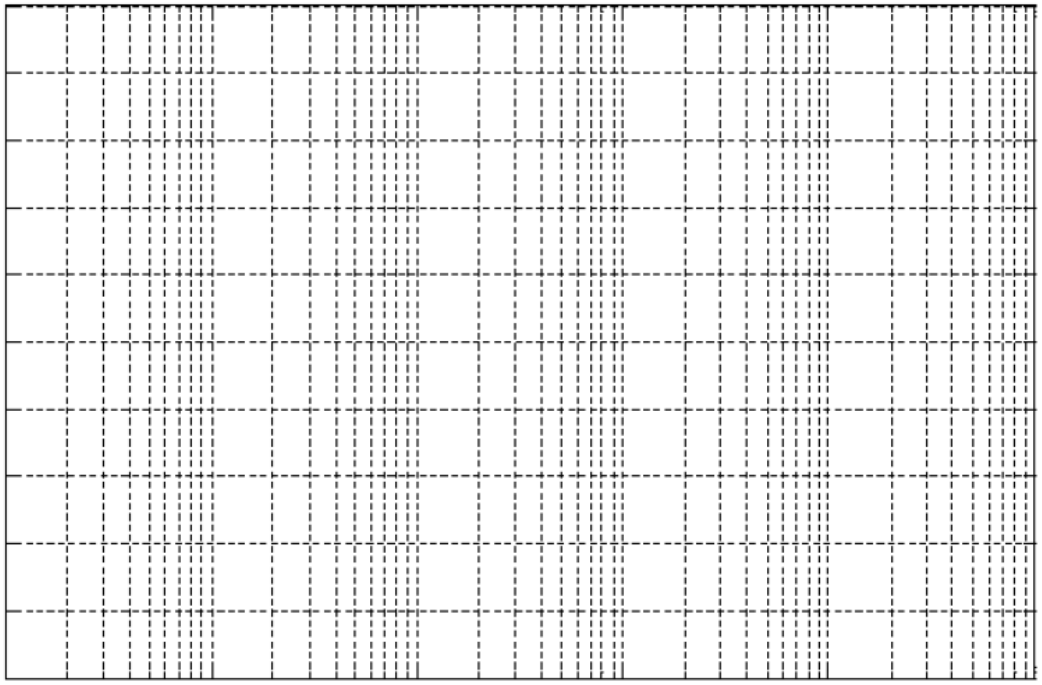
Section B: Bode Plot Approximation

Exercise B

Sketch the Bode plot for the transfer functions in Exercise A by hand without the help of MATLAB. Use the clean Bode plot provided in Appendix 1 below. Show the analytical steps involved in sketching the plots.

Appendix 1

Bode Plot



Bode Plot

